Amazon Sales Analysis

Final Detailed Analysis Report By Sourav Roychoudhury

Python | Pandas | NumPy | Matplotlib | Seaborn

Executive Summary

This project analyzes an Amazon Sales dataset using Python to uncover sales trends, profitability, customer purchasing patterns, operational efficiency, and regional performance. The analysis covers data cleaning, exploratory data analysis (EDA), statistical analysis, correlation analysis, and business intelligence reporting.

The objective is to help decision-makers understand:

- Which regions generate the highest revenue
- Which products are the most profitable
- How sales channels affect operations
- Seasonal purchasing behavior
- Shipping performance and customer buying trends

Key findings: Sub-Saharan Africa contributes the highest sales revenue, Djibouti generates the highest profit, Cosmetics is the most profitable product category, while Fruits contributes the least revenue and profit. The analysis also reveals a moderate positive relationship between product price, units sold, and overall profit.

1. Project Objective

The primary objective was to analyze Amazon sales transactions to generate actionable business insights by examining:

- Regional sales performance
- Country-wise profitability
- Product category performance
- Sales channel comparison
- Order processing efficiency
- Revenue trends over time
- Correlation between business variables

2. Dataset Overview

The dataset contains transactional information representing international sales across multiple countries and product categories. Key columns include:

- Region, Country
- Item Type, Sales Channel, Order Priority
- Order Date, Ship Date
- Units Sold, Unit Price, Unit Cost
- Total Revenue, Total Cost, Total Profit

3. Data Cleaning & Preprocessing

Date Conversion

Order Date and Ship Date were converted into datetime format for time-based analysis.

Missing Value Analysis

- Missing values were checked using .isnull() and heatmap visualization
- A sample missing value was intentionally introduced for demonstration
- Missing numerical values were replaced using the mean

Data Type Verification

Relevant columns were converted into appropriate numerical and datetime formats.

Statistical Summary

Descriptive statistics were generated including mean, median, standard deviation, minimum, maximum, and quartiles.

4. Exploratory Data Analysis

A. Regional Revenue Analysis

Sub-Saharan Africa recorded the highest total sales revenue, indicating strong market demand, high purchasing activity, and successful regional operations. Recommendation: increase inventory, marketing budget, and product availability for this region.

B. Average Product Pricing

Product Category	Observation
Household	Highest average cost among most categories
Cosmetics	Very high selling price
Fruits	Lowest unit price
Baby Food	High pricing with healthy margins

High-priced products generally generate larger profit margins, especially Cosmetics and Household items.

C. Country-wise Profit Analysis

Djibouti produces the highest profit despite not necessarily having the largest order count, suggesting premium products, better margins, and high-value transactions.

5. Sales Channel Analysis

The dataset contains two sales channels: Online and Offline.

Order Priority Distribution

Priority	Offline Orders	Online Orders
High	17	13
Critical	13	9
Low	12	15
Medium	8	13

Offline orders show a slightly greater proportion of high-priority transactions, while online orders are more evenly distributed across priorities.

6. Order Processing Time

Sales Channel	Average Processing Time
Offline	~23 days 4 hours
Online	~23 days 12 hours

There is very little difference between online and offline fulfillment speed, indicating efficient logistics and a consistent shipping process.

7. Item Type Performance

Cosmetics recorded the highest revenue due to higher selling prices, better margins, and strong customer demand. Fruits recorded the lowest revenue due to low pricing and smaller contribution per unit.

8. Correlation Analysis

Metric	Correlation Coefficient	Interpretation
Unit Price vs Total Profit	0.557	Moderate positive — higher-priced products generally earn more profit
Units Sold vs Total Profit	0.565	Moderate positive — higher volume generally increases profit

Profitability depends on both pricing and volume, but also on product margins and production costs.

9. Seasonal Sales Trends

Monthly revenue analysis revealed noticeable seasonal fluctuations.

- Peak months: February, November, April
- Lower revenue months: August, March

Marketing campaigns should focus on high-demand months to maximize revenue, while inventory can be optimized during slower periods.

10. Additional Findings

Sales Channel Pricing

Sales Channel	Average Unit Price
Offline	$310.72
Online	$242.80

Offline transactions tend to involve higher-priced products, possibly due to bulk purchasing, enterprise customers, or premium product sales.

Revenue by Age Group

Region	Revenue Ranking
Sub-Saharan Africa	Highest total revenue
Asia	Highest average revenue per order

Profit by Product Category

Category	Performance
Cosmetics	Highest profit
Household	Strong performer
Office Supplies	Strong performer
Clothes	Strong performer
Fruits	Lowest profit

Outlier Detection

Outliers were detected in Total Cost using the Interquartile Range (IQR) method. These may represent bulk orders, high-value purchases, or exceptional transactions requiring further investigation.

Key Business Insights

- Sub-Saharan Africa is the strongest revenue-generating region
- Djibouti contributes the highest total profit
- Cosmetics is the most profitable and highest-revenue product category
- Fruits contribute the least revenue and profit
- Offline transactions involve higher-priced products on average
- Shipping performance is consistent across both sales channels
- February, November, and April are peak sales periods
- Both unit price and units sold show moderate positive relationships with total profit
- Outlier detection identifies potential bulk or high-value transactions for further review

Business Recommendations

1. Expand in High-Performing Regions: Increase marketing, inventory, and operational investment in Sub-Saharan Africa and Asia to capitalize on strong demand.
2. Prioritize High-Margin Products: Focus promotions and inventory on Cosmetics, Household items, and Office Supplies, which consistently deliver higher revenue and profit.

3. Optimize Underperforming Categories: Review pricing, sourcing, and promotional strategies for Fruits and other low-margin products to improve profitability.
4. Leverage Seasonal Demand: Align inventory planning and marketing campaigns with peak revenue months (February, November, and April) to maximize sales opportunities.
5. Enhance Country-Specific Strategies: Develop targeted sales strategies for high-profit countries such as Djibouti while improving logistics in slower-performing regions.
6. Maintain Logistics Efficiency: Continue monitoring shipping performance, as online and offline channels currently exhibit similar processing times.
7. Monitor High-Value Transactions: Investigate outliers in Total Cost to identify bulk orders, premium customers, or unusual purchasing behavior.
8. Increase Revenue Through Product Mix: Encourage sales of higher-priced, higher-margin products in both online and offline channels to improve overall profitability.

Conclusion

This Amazon Sales Analysis demonstrates how data analytics can transform raw transactional data into meaningful business intelligence. Through comprehensive exploratory data analysis, statistical evaluation, and visualization, the project identifies the strongest markets, most profitable product categories, seasonal demand patterns, and operational efficiencies.

The findings provide actionable recommendations that can help improve revenue growth, inventory planning, marketing effectiveness, and supply chain performance. Overall, the project showcases practical applications of Python-based data analysis for supporting strategic business decision-making and serves as a strong portfolio project for data analytics roles.